A new architecture must go through almost same level of scrutiny and audits as a drug goes through trials before it is released in market for public.

x86 or A64, how to move forward?

It is very clear that there is a shift from x86-based architecture. To get higher performances from a supercomputer, most modern-day supercomputers are adopting ARM's architecture or using AMD's Zen architecture.

Zen architecture from AMD is also a closed system architecture, which means that no one other than AMD can develop a high-performance silicon-based compute on Zen architecture.

ARM's Neoverse architecture is the only leading architecture available for use to develop HPC silicon processor. With an under development ecosystem of HPC software stacks for ARM's A64-based architecture, it is not a question if an ARM-based server will become mainstream, but the question is how soon.

Europe's SiPearl has also developed an ARM-based high-performance compute processor 'Rhea.' The seed funding for Rhea came from the European Union. In Japan, Fugaku was funded by public funds. Similar programmes are running in USA too.

India is growing at a very fast pace and the need for data processing on HPC machines will exponentially increase in the next 5-10 years. Almost all government institutions will need HPC servers of their own in future. Each processor costs in range of $2000-$3000 and every HPC server requires minimum of four such processors.

India must continue designing and manufacturing Intel x86-based supercomputers and HPC machines. However, in parallel, there is a need to have India's own HPC silicon processor to cater to needs of organisations in India. ARM-based architecture provides a window for India to design, develop, deploy, and export HPC servers. With indigenously developed processor, the cost could be brought down to $1500-$2000, which is a cost benefit of ~40%.

A multi-pronged approach towards supercomputer architecture is needed to keep pace with the world and reduce dependency on established players and countries. END

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